Appendix B — Index to Notations

Appendix B. Index to Notations

In the following formulas, letters that are not further qualified have the following significance:

j, k	integer-valued arithmetic expression
m, n	nonnegative integer-valued arithmetic expression
p, q	binary-valued arithmetic expression (0 or 1)
x, y	real-valued arithmetic expression
z	complex-valued arithmetic expression
f	integer-valued, real-valued, or complex-valued function
G, H	graph
S, T	set or multiset
$\mathcal{F}, \mathcal{G}$	family of sets
u, v	vertex of a graph
α, β	string of symbols

The place of definition is either a page number in the present volume or a section number in another volume. Many other notations, such as K_n for the complete graph on n vertices, appear in the main index at the close of this book.

Formal symbolism	Meaning	Where defined
$V \leftarrow E$	give variable V the value of expression E	§1.1
$U \leftrightarrow V$	interchange the values of variables U and V	§1.1
A_n or $A[n]$	the nth element of linear array A	§1.1
A_{mn} or $A[m,n]$	the element in row m and column n of rectangular array A	§1.1
$(R?\ a\colon b)$	conditional expression: denotes a if relation R is true, b if R is false	96
$[R]$	characteristic function of relation R: $(R?\ 1\colon 0)$	§1.2.3
δ_{jk}	Kronecker delta: $[j=k]$	§1.2.3
$[z^n]\, f(z)$	coefficient of z^n in power series $f(z)$	§1.2.9
$z_1 + z_2 + \cdots + z_n$	sum of n numbers (even when n is 0 or 1)	§1.2.3
$a_1 a_2 \ldots a_n$	product or string or vector of n elements	
$(x_1, \ldots, x_n)$	vector of n elements	
$\langle x_1 x_2 \ldots x_{2k-1} \rangle$	median value (middle value after sorting)	75

$\sum_{R(k)} f(k)$	sum of all $f(k)$ such that relation $R(k)$ is true	§1.2.3
$\prod_{R(k)} f(k)$	product of all $f(k)$ such that relation $R(k)$ is true	§1.2.3
$\min_{R(k)} f(k)$	minimum of all $f(k)$ such that relation $R(k)$ is true	§1.2.3
$\max_{R(k)} f(k)$	maximum of all $f(k)$ such that relation $R(k)$ is true	§1.2.3
$\bigcup_{R(k)} S(k)$	union of all $S(k)$ such that relation $R(k)$ is true	
$\sum_{k=a}^{b} f(k)$	shorthand for $\sum_{a\le k\le b} f(k)$	§1.2.3
$\{a \mid R(a)\}$	set of all a such that relation $R(a)$ is true	
$\sum\{f(k) \mid R(k)\}$	another way to write $\sum_{R(k)} f(k)$	
$\{a_1, a_2, \ldots, a_n\}$	the set or multiset $\{a_k \mid 1 \le k \le n\}$	
$[x\,.\,.\,y]$	closed interval: $\{a \mid x \le a \le y\}$	§1.2.2
$(x\,.\,.\,y)$	open interval: $\{a \mid x < a < y\}$	§1.2.2
$[x\,.\,.\,y)$	half-open interval: $\{a \mid x \le a < y\}$	§1.2.2
$(x\,.\,.\,y]$	half-closed interval: $\{a \mid x < a \le y\}$	§1.2.2
$\lvert S\rvert$	cardinality: the number of elements in S	
$\lvert f\rvert$	number of solutions (when f is Boolean): $\sum_x f(x)$	207
$\lvert x\rvert$	absolute value of x: $(x \ge 0?\ x\colon -x)$	
$\lvert z\rvert$	absolute value of z: $\sqrt{z\bar{z}}$	§1.2.2
$\lvert\alpha\rvert$	length of α: m if $\alpha = a_1a_2\ldots a_m$	
$\lfloor x\rfloor$	floor of x, greatest integer function: $\max_{k\le x} k$	§1.2.4
$\lceil x\rceil$	ceiling of x, least integer function: $\min_{k\ge x} k$	§1.2.4
$x \bmod y$	mod function: $\bigl(y = 0?\ x\colon x - y\lfloor x/y\rfloor\bigr)$	§1.2.4
$\{x\}$	fractional part (used in contexts where a real value, not a set, is implied): $x \bmod 1$	§1.2.11.2
$x \equiv x' \pmod{y}$	relation of congruence: $x \bmod y = x' \bmod y$	§1.2.4
$j\backslash k$	j divides k: $k \bmod j = 0$ and $j > 0$	§1.2.4
$S \setminus T$	set difference: $\{s \mid s \text{ in } S \text{ and } s \text{ not in } T\}$	
$S \setminus t$	shorthand for $S \setminus \{t\}$	
$G \setminus U$	G with vertices of the set U removed	13
$G \setminus v$	G with vertex v removed	13
$G \setminus e$	G with edge e removed	13

$G\,/\,e$	G with edge e shrunk to a point	463
$S \cup t$	shorthand for $S \cup \{t\}$	
$S \uplus T$	multiset sum; e.g., $\{a, b\} \uplus \{a, c\} = \{a, a, b, c\}$	§4.6.3
$\gcd(j, k)$	greatest common divisor: $(j{=}k{=}0?\ 0\colon \max_{d\backslash j, d\backslash k} d)$	§1.1
$j \perp k$	j is relatively prime to k: $\gcd(j, k) = 1$	§1.2.4

A^T	transpose of rectangular array A: $A^T[j,k] = A[k,j]$	
α^R	left-right reversal of string α	
α^T	conjugate of partition α	394
x^y	x to the y power (when $x > 0$): $e^{y \ln x}$	§1.2.2
x^k	x to the k power: $\bigl(k \ge 0?\ \prod_{j=0}^{k-1} x\colon 1/x^{-k}\bigr)$	§1.2.2
x^-	inverse (or reciprocal) of x: x^{-1}	§1.3.3
$x^{\overline{k}}$	x to the k rising: $\Gamma(x+k)/\Gamma(x) =$ $\bigl(k \ge 0?\ \prod_{j=0}^{k-1}(x+j)\colon 1/(x+k)^{\overline{-k}}\bigr)$	§1.2.5
$x^{\underline{k}}$	x to the k falling: $x!/(x-k)! =$ $\bigl(k \ge 0?\ \prod_{j=0}^{k-1}(x-j)\colon 1/(x-k)^{\underline{-k}}\bigr)$	§1.2.5
$n!$	n factorial: $\Gamma(n+1) = n^{\underline{n}}$	§1.2.5
$\binom{x}{k}$	binomial coefficient: $(k < 0?\ 0\colon x^{\underline{k}}/k!)$	§1.2.6
$\binom{n}{n_1,\ldots,n_m}$	multinomial coefficient (when $n = n_1 + \cdots + n_m$)	§1.2.6
${n \brack m}$	Stirling cycle number: $\sum_{0<k_1<\cdots<k_{n-m}<n} k_1 \ldots k_{n-m}$	§1.2.6
${n \brace m}$	Stirling partition number: $\sum_{1 \le k_1 \le \cdots \le k_{n-m} \le m} k_1 \ldots k_{n-m}$	§1.2.6
$\left\langle {n \atop m} \right\rangle$	Eulerian number: $\sum_{k=0}^{m}(-1)^k \binom{n+1}{k}(m+1-k)^n$	§5.1.3
$\left\lvert {n \atop m} \right\rvert$	m-part partitions of n: $\sum_{1 \le k_1 \le \cdots \le k_m} [k_1 + \cdots + k_m = n]$	399
$(\ldots a_1 a_0 . a_{-1} \ldots)_b$	radix-b positional notation: $\sum_k a_k b^k$	§4.1
$\Re z$	real part of z	§1.2.2
$\Im z$	imaginary part of z	§1.2.2
$\overline{z}$	complex conjugate: $\Re z - i\,\Im z$	§1.2.2
$\neg p$ or $\sim p$ or $\overline{p}$	complement: $1 - p$	49
$\sim x$ or $\overline{x}$	bitwise complement	135
$p \wedge q$	Boolean conjunction (and): pq	49
$x \wedge y$	minimum: $\min\{x, y\}$	63
$x \mathbin{\&} y$	bitwise AND	134
$p \vee q$	Boolean disjunction (or): $\overline{\overline{p}\,\overline{q}}$	49
$x \vee y$	maximum: $\max\{x, y\}$	63
$x \mid y$	bitwise OR	134
$p \oplus q$	Boolean exclusive disjunction (xor): $(p+q) \bmod 2$	50

$p \oplus q$	Boolean exclusive disjunction (xor): $(p + q) \bmod 2$	[illegible]
$x \oplus y$	bitwise XOR	134
$x \dot{-} y$	saturating subtraction, x monus y: $\max\{0, x - y\}$	§1.3.1′
$x \ll k$	bitwise left shift: $\lfloor 2^k x \rfloor$	135
$x \gg k$	bitwise right shift: $x \ll (-k)$	135
$x \ddagger y$	"zipper function" for interleaving bits, x zip y	147

$\log_b x$	logarithm, base b, of x (defined when $x > 0$, $b > 0$, and $b \neq 1$): the y such that $x = b^y$	§1.2.2
$\ln x$	natural logarithm: $\log_e x$	§1.2.2
$\lg x$	binary logarithm: $\log_2 x$	§1.2.2
λn	binary logsize (when $n > 0$): $\lfloor \lg n \rfloor$	142
$\exp x$	exponential of x: $e^x = \sum_{k=0}^{\infty} x^k/k!$	§1.2.9
ρn	ruler function (when $n > 0$): $\max_{2^m \backslash n} m$	140
νn	sideways sum (when $n \geq 0$): $\sum_{k \geq 0} \bigl((n \gg k) \mathbin{\&} 1\bigr)$	143
$\langle X_n \rangle$	the infinite sequence $X_0, X_1, X_2, \ldots$ (here the letter n is part of the symbolism)	§1.2.9
$f'(x)$	derivative of f at x	§1.2.9
$f''(x)$	second derivative of f at x	§1.2.10
$H_n^{(x)}$	harmonic number of order x: $\sum_{k=1}^{n} 1/k^x$	§1.2.7
H_n	harmonic number: $H_n^{(1)}$	§1.2.7
F_n	Fibonacci number: $(n \leq 1?\ n\colon F_{n-1} + F_{n-2})$	§1.2.8
B_n	Bernoulli number: $n!\,[z^n]\,z/(1 - e^{-z})$	§1.2.11.2
$\det(A)$	determinant of square matrix A	§1.2.3
$\operatorname{sign}(x)$	sign of x: $[x > 0] - [x < 0]$	
$\zeta(x)$	zeta function: $\lim_{n\to\infty} H_n^{(x)}$ (when $x > 1$)	§1.2.7
$\Gamma(x)$	gamma function: $(x-1)! = \gamma(x, \infty)$	§1.2.5
$\gamma(x, y)$	incomplete gamma function: $\int_0^y e^{-t} t^{x-1}\,dt$	§1.2.11.3
γ	Euler's constant: $-\Gamma'(1) = \lim_{n\to\infty}(H_n - \ln n)$	§1.2.7
e	base of natural logarithms: $\sum_{n\geq 0} 1/n!$	§1.2.2
π	circle ratio: $4\sum_{n\geq 0}(-1)^n/(2n+1)$	§1.2.2
∞	infinity: larger than any number	
Λ	null link (pointer to no address)	§2.1
$\emptyset$	empty set (set with no elements)	
ϵ	empty string (string of length zero)	
ϵ	unit family: $\{\emptyset\}$	273
ϕ	golden ratio: $(1 + \sqrt{5})/2$	§1.2.8
$\varphi(n)$	Euler's totient function: $\sum_{k=0}^{n-1} [k \perp n]$	§1.2.4
$x \approx y$	x is approximately equal to y	§1.2.5

$x \sim y$	x is approximately equal to y	§1.2.5
$G \cong H$	G is isomorphic to H	14
$O\big(f(n)\big)$	big-oh of $f(n)$, as the variable $n \to \infty$	§1.2.11.1
$O\big(f(z)\big)$	big-oh of $f(z)$, as the variable $z \to 0$	§1.2.11.1
$\Omega\big(f(n)\big)$	big-omega of $f(n)$, as the variable $n \to \infty$	§1.2.11.1
$\Theta\big(f(n)\big)$	big-theta of $f(n)$, as the variable $n \to \infty$	§1.2.11.1

$\overline{G}$	complement of graph (or uniform hypergraph) G	26
G^T	converse of digraph G (change '$\longrightarrow$' to '$\longleftarrow$')	§7.2.2.3
$G \mid U$	G restricted to the vertices of set U	13
$u \longrel v$	u is adjacent to v	13
$u \not\!\!\longrel v$	u is not adjacent to v	13
$u \longrightarrow v$	there is an arc from u to v	18
$u \longrightarrow^* v$	transitive closure: v is reachable from u	159
$d(u,v)$	distance from u to v	16
$G \cup H$	union of G and H	26
$G \oplus H$	direct sum (juxtaposition) of G and H	26
$G \longrel H$	join of G and H	26
$G \longrightarrow H$	directed join of G and H	26
$G \,\square\, H$	Cartesian product of G and H	27
$G \otimes H$	direct product (conjunction) of G and H	28
$G \boxtimes H$	strong product of G and H	28
$G \vartriangle H$	odd product of G and H	28
$G \circ H$	lexicographic product (composition) of G and H	28
e_j	elementary family: $\{\{j\}\}$	273
$\wp$	universal family: all subsets of a given universe	275
$\mathcal{F} \cup \mathcal{G}$	union of families: $\{S \mid S \in \mathcal{F}$ or $S \in \mathcal{G}\}$	273
$\mathcal{F} \cap \mathcal{G}$	intersection of families: $\{S \mid S \in \mathcal{F}$ and $S \in \mathcal{G}\}$	273
$\mathcal{F} \setminus \mathcal{G}$	difference of families: $\{S \mid S \in \mathcal{F}$ and $S \notin \mathcal{G}\}$	273
$\mathcal{F} \oplus \mathcal{G}$	symmetric difference of families: $(\mathcal{F} \setminus \mathcal{G}) \cup (\mathcal{G} \setminus \mathcal{F})$	273
$\mathcal{F} \sqcup \mathcal{G}$	join of families: $\{S \cup T \mid S \in \mathcal{F}, T \in \mathcal{G}\}$	273
$\mathcal{F} \sqcap \mathcal{G}$	meet of families: $\{S \cap T \mid S \in \mathcal{F}, T \in \mathcal{G}\}$	273
$\mathcal{F} \boxplus \mathcal{G}$	delta of families: $\{S \oplus T \mid S \in \mathcal{F}, T \in \mathcal{G}\}$	273
$\mathcal{F}/\mathcal{G}$	quotient (cofactor) of families	273
$\mathcal{F} \bmod \mathcal{G}$	remainder of families: $\mathcal{F} \setminus (\mathcal{G} \sqcup (\mathcal{F}/\mathcal{G}))$	273
$\mathcal{F} \mathbin{\S} k$	symmetrized family, if $\mathcal{F} = e_{j_1} \cup e_{j_2} \cup \cdots \cup e_{j_n}$	274
$\mathcal{F}^{\uparrow}$	maximal elements of $\mathcal{F}$: $\{S \in \mathcal{F} \mid T \in \mathcal{F}$ and $S \subseteq T$ implies $S = T\}$	276
$\mathcal{F}^{\downarrow}$	minimal elements of $\mathcal{F}$	

$\mathcal{F}^{\downarrow}$	minimal elements of $\mathcal{F}$: $\{S \in \mathcal{F} \mid T \in \mathcal{F} \text{ and } S \supseteq T \text{ implies } S = T\}$	276
$\mathcal{F} \nearrow \mathcal{G}$	nonsubsets: $\{S \in \mathcal{F} \mid T \in \mathcal{G} \text{ implies } S \not\subseteq T\}$	276
$\mathcal{F} \searrow \mathcal{G}$	nonsupersets: $\{S \in \mathcal{F} \mid T \in \mathcal{G} \text{ implies } S \not\supseteq T\}$	276
$\mathcal{F} \swarrow \mathcal{G}$	subsets: $\{S \in \mathcal{F} \mid T \in \mathcal{G} \text{ implies } S \subseteq T\} = \mathcal{F} \setminus (\mathcal{F} \nearrow \mathcal{G})$	669
$\mathcal{F} \nwarrow \mathcal{G}$	supersets: $\{S \in \mathcal{F} \mid T \in \mathcal{G} \text{ implies } S \supseteq T\} = \mathcal{F} \setminus (\mathcal{F} \searrow \mathcal{G})$	669
$X \cdot Y$	dot product of vectors: $x_1y_1 + x_2y_2 + \cdots + x_ny_n$, if $X = x_1x_2\ldots x_n$ and $Y = y_1y_2\ldots y_n$	12
$X \subseteq Y$	containment of vectors: $x_k \le y_k$ for $1 \le k \le n$, if $X = x_1x_2\ldots x_n$ and $Y = y_1y_2\ldots y_n$	135
$x \subseteq y$	bitwise containment: $x_k \le y_k$ for all k, if $x = (\ldots x_1x_0.x_{-1}\ldots)_2$, $y = (\ldots y_1y_0.y_{-1}\ldots)_2$	135
$\alpha \diamond \beta$	melding of truth tables	218
$\alpha(G)$	independence number of G	35
$\gamma(G)$	domination number of G	673
$\kappa(G)$	vertex connectivity of G	§7.4.1.3
$\lambda(G)$	edge connectivity of G	§7.4.1.3
$\nu(G)$	matching number of G	§7.5.5
$\chi(G)$	chromatic number of G	35
$\omega(G)$	clique number of G	35
$c(G)$	number of spanning trees of G	482
▮	end of algorithm, program, or proof	§1.1

And to auoide the tediouse repetition of these woordes : is equalle to :
I will sette as I doe often in woorke use, a paire of paralleles,
or Gemowe lines of one lengthe, thus: ====,
bicause noe .2. thynges, can be moare equalle.

— ROBERT RECORDE, *The Whetstone of Witte* (1557)

Prof. Le Gendre, in the treatise that we shall often have occasion to cite,
used the same sign for both equality and congruence.
To avoid ambiguity we have made a distinction.

— C. F. GAUSS, *Disquisitiones Arithmeticæ* (1801)

Someone told me that each equation I included in the book
would halve the sales.

— STEPHEN HAWKING, *A Brief History of Time* (1987)

A reader who has temporarily forgotten the definition of, say, "*z*"

in an argument can at least guess that it should be a complex number.

—TERENCE TAO, *Google Buzz feed* (20 March 2010)